9. Develop an Android application implementing Static and Dynamic Broadcast Receivers.

Project Name: BroadcastDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center">

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Broadcast Receiver Example"

       android:textSize="22sp"/>

</LinearLayout>
```

Create a Broadcast Receiver Class.

Right Click:-> java package → New → Java Class Class Name: MyReceiver

```
package com.example.broadcastdemo;

import android.content.BroadcastReceiver;

import android.content.Context;

import android.content.Intent;

import android.widget.Toast;


public class MyReceiver extends BroadcastReceiver {


   @Override

   public void onReceive(Context context, Intent intent) {

       Toast.makeText(

               context,

               "Broadcast Received",

               Toast.LENGTH_SHORT

       ).show();

   }

}
```

Static Broadcast Receiver.

Open:AndroidManifest.xml

Inside: <application>

Add:

```
<receiver
   android:name=".MyReceiver"
   android:exported="true">
   <intent-filter>
       <action android:name="android.intent.action.AIRPLANE_MODE" />
   </intent-filter>
</receiver>
```

```
        android:supportsRtl="true"
        android:theme="@style/Theme.BroadcastDemo">

        <receiver
            android:name=".MyReceiver"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.AIRPLANE_MODE" />
            </intent-filter>
        </receiver>

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
```

Dynamic Broadcast Receiver.

File Name: MainActivity.java

```
package com.example.broadcastdemo;

import android.content.IntentFilter;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


   MyReceiver myReceiver = new MyReceiver();


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       // Dynamic Receiver

       IntentFilter filter = new IntentFilter();

       filter.addAction("android.intent.action.AIRPLANE_MODE");


       ContextCompat.registerReceiver(

               this,

               myReceiver,

               filter,

               ContextCompat.RECEIVER_EXPORTED
```

```
        );
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        unregisterReceiver(myReceiver);
    }
}
```

Turn on airplane mode

